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DSPD 2783 - CURRENT TOPICS IN COMPUTER SCIENCE

**The Ethical Issues of User Data Collection on Social Media
Platforms**

CASE STUDY REPORT

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**The Ethical Issues of User Data Collection on Social Media
Platforms**

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**A report submitted in fulfilment of the requirements
for the award of the
Diploma in Computer Science**

School of Professional and Continuing Education

School of Professional and Continuing Education

Universiti Teknologi Malaysia

February 2026

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ABSTRACT

This paper explores ethical implications of data privacy associated with social media applications, as data privacy has become an integral element in today's computing world. Social media applications facilitate the means through which billions of individuals globally connect; however, data privacy issues related to extensive user data collected, stored and utilized by social media applications have raised considerable concerns when it comes to privacy, security, fairness and responsibility. One significant finding of this study is that a lack of user knowledge as to what constitutes personal information and how their personal data is collected, managed, and shared with third parties is a major factor contributing to the above-mentioned privacy issues.

Through qualitative methodology (literature review, examination of recent incidents involving privacy), this study investigates ethical issues associated with social media applications, with the specific focus on the investigation placed on four major ethical dimensions (Privacy, Security, Justice and Accountability). Results indicate that data privacy issues are primarily driven by data-based business models, insufficient transparency and regulatory enforcement, and limited number of users with an understanding of privacy policies, and can result in a variety of negative consequences such as identity theft, data-sharing with third parties without consent, reputational harm, financial liability and diminished public trust in digital platforms.

The study also explores the effects which data privacy concerns have on individuals, organizations and society as a whole. From these findings, several recommendations are made including: stronger

security measure; more stringent privacy regulations; increased accountability for organizations using personal data; and increasing user knowledge about how to protect the privacy of their data. Ultimately, it concludes that ethical management of data & responsible governance are necessary for social media technology to continue to provide benefits to society and at the same time protect user rights and digital security.

ABSTRAK

Kertas kerja ini meneroka implikasi etika privasi data yang dikaitkan dengan aplikasi media sosial, kerana privasi data telah menjadi elemen penting dalam dunia pengkomputeran hari ini. Aplikasi media sosial memudahkan cara berbilion-bilion individu berhubung secara global; walau bagaimanapun, isu privasi data yang berkaitan dengan data pengguna yang banyak yang dikumpul, disimpan dan digunakan oleh aplikasi media sosial telah menimbulkan kebimbangan besar apabila ia berkaitan dengan privasi, keselamatan, keadilan dan tanggungjawab. Satu penemuan penting kajian ini ialah kekurangan pengetahuan pengguna tentang perkara yang membentuk maklumat peribadi dan cara data peribadi mereka dikumpul, diurus dan dikongsi dengan pihak ketiga merupakan faktor utama yang menyumbang kepada isu privasi yang dinyatakan di atas.

Melalui metodologi kualitatif (semakan literatur, pemeriksaan insiden terkini yang melibatkan privasi), kajian ini menyiasat isu etika yang berkaitan dengan aplikasi media sosial, dengan tumpuan khusus pada penyiasatan diletakkan pada empat dimensi etika utama (Privasi, Keselamatan, Keadilan dan Akauntabiliti). Keputusan menunjukkan bahawa isu privasi data didorong terutamanya oleh model perniagaan berasaskan data, ketelusan dan penguatkuasaan kawal selia yang tidak mencukupi, dan bilangan pengguna yang terhad dengan pemahaman tentang dasar privasi, dan boleh mengakibatkan pelbagai akibat negatif seperti kecurian identiti, perkongsian data dengan pihak ketiga tanpa kebenaran, kemudaratan reputasi, liabiliti kewangan dan kepercayaan awam yang berkurangan terhadap platform digital.

Kajian ini juga meneroka kesan kebimbangan privasi data terhadap individu, organisasi dan masyarakat secara keseluruhan. Daripada penemuan ini, beberapa cadangan dibuat termasuk: langkah keselamatan yang lebih kukuh; peraturan privasi yang lebih ketat; meningkatkan akauntabiliti untuk organisasi yang menggunakan data peribadi; dan meningkatkan pengetahuan pengguna tentang cara melindungi privasi data mereka. Akhirnya, ia menyimpulkan bahawa pengurusan data yang beretika & tadbir urus yang bertanggungjawab adalah perlu untuk teknologi media sosial terus memberi manfaat kepada masyarakat dan pada masa yang sama melindungi hak pengguna dan keselamatan digital.

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
API	-	Application Programming Interface
EU	-	European Union
GDPR	-	General Data Protection Regulation
ICT	-	Information and Communication Technology
IT	-	Information Technology
PDPA	-	Personal Data Protection Act
SNS	-	Social Networking Sites
UI	-	User Interface
URL	-	Uniform Resource Locator
US	-	United States
VPN	-	Virtual Private Network

CHAPTER 1 INTRODUCTION & BACKGROUND

1.1 Background of Social Media Platforms

When talking about communication in today's society, social media plays a major role in that process for people worldwide. For example, you can use Insta, Facebook, Tiktok, X (Formally Twitter), and LinkedIn to find out about all sorts of different things, communicate with others, and access varying types of Digital Content. With the increased amount of time people are spending using the internet plus the increased number of people that use mobile technologies, the number of social media users continues to grow rapidly around the world. And as these technologies continue to change so do the amounts of personal data collected and processed by these companies to improve how people use these sites and offer much better individual user-specific services.

1.2 Importance of Data Privacy in the Digital Era

Today most people are worried about their own personal data. Social media sites gather a lot of information about us like our names, where we live, what we look at online, what we like and who we talk to. This data helps sites show us ads and content that interest us. It also makes us vulnerable to hacking, identity theft and other problems. We need to keep users personal data to build and ensure digital practices are fair and secure. Protecting data is important for individuals, organizations and governments. Social media platforms collect data to deliver content and advertisements but data privacy is a critical concern. People organizations and governments must work together to maintain trust, security and ethical digital practices. Lastly, data privacy is important to prevent access, data breaches and misuse of personal information

1.3 Problem Statement

Social media platforms have a lot of benefits. Many people who use them do not know what happened to them if they overshared personal information. There have been cases where this information has been stolen or used wrongly. This is a problem because it shows that the ways we currently protect people's privacy are not working very well. When people sign up for media they usually agree to the rules without really understanding what they mean. This can put their information at risk. We need to take a look at how social media platforms handle people's private information and figure out ways to help people understand what is going on with social media and how to keep their information safe with social media.

1.4 Research Goal

This study will focus on exploring the issues surrounding data privacy concerns with respect to social media sites; it will also look into various ethical matters stemming from these issues; finally, it will examine ways to keep one's private information safe when using an online site.

1.4.1 Research Objectives

1. To examine data privacy issues in social media platforms.
2. To analyze privacy concerns from ethical perspectives.
3. To identify the root causes of privacy violations.
4. To evaluate impacts on individuals, organizations, and society.
5. To propose realistic solutions.

1.5 Scope of the Study

This study is about the problems with keeping information private on social media platforms like Facebook, Instagram, Tiktok and Twitter (X). It looks at how these social media platforms collect, use and protect the information of users. It also discusses user awareness of privacy settings and the potential risks associated with sharing personal information online. This study deals with people who use social media and does not look at privacy issues with other online services like online banking buying things online or healthcare systems, on the internet.

CHAPTER 2 LITERATURE REVIEW

2.1 Introduction

The increase of social media sites in our society over the years has made it easier for individuals to connect and communicate. Sites like Facebook, Instagram, TikTok, X (formerly Twitter), and LinkedIn are a major resource for many individuals globally. Although it is easy to communicate, chat with friends, family members, and new individuals, social media sites are collecting large amounts of your personal data. This data includes such things as your name, location, email address, browsing history, and other types of information about your online activities. The amount of information collected about an individual has become concerning when it comes to the privacy and security of this data as well as ethical responsibilities by social media companies.

In recent years, researchers have begun to examine how social media companies gather, analyze, and use users' personal information. Many studies identify the advantages of social media companies gathering user information, including customizing services, delivering targeted ads, and enhancing users' overall experience as a result of the information collected. However, researchers have also identified possible threats associated with social media companies using personal information, including unauthorised access to personal information, misuse of information, data breaches, and lack of transparency. As a result, the Identification of these potential effects has led to the establishment of laws and policies by governments and other organizations to protect the privacy of users.

This section examines the existing research regarding personal data collection and social media. This section covers major concepts of privacy, ethical issues

relating to how social media is used, important privacy violation cases recently that occurred, and limitations of previous research. This literature review contains the essential elements about the key concepts, ethical issues and real-world examples of data privacy on social media. The literature review builds a basis to explain the causes, effects and ethical issues associated with the privacy problems discussed in subsequent chapters. A stronger overall knowledge and understanding of the current status and issues in the area will be achieved with this review.

2.2 Social Media and Personal Data Collection

The collection of data regarding an individual's social media activity supports the goals of many social network providers. Each time someone likes or shares a message, adds comments, follows other users or clicks on advertisements, they generate information which can be of tremendous value to the organisation providing their services. Social networks will utilise this data to build user profiles, thereby developing a better understanding of how users access and engage with the content presented on the social network. Data collected through these behaviours are often used in creating targeted ads and personalising recommendations and content served to users.

Researchers have found that when you set up a new social media profile, you often share your personal details, such as your full name, date of birth, email address, phone number, pictures of yourself and where you live. In addition to your voluntary contributions, social media companies can obtain additional information from you using cookies, tracking technologies and device identification software. Through these methods, social media companies are able to monitor your activities on and off their site.

Data collection can create positive outcomes for users, including an enhanced user experience as well as more tailored content served to them. On the flip side, these activities present a potential threat to your privacy. Many users do not realize how much information they have provided to social media companies or how the information is being utilized. Many privacy policies are very long and contain legal jargon, which makes it extremely difficult for them to provide informed consent. In addition, some users may have their information shared with third parties due to an advertising alliance agreement or business partnership.

The argument has been made by researchers that due to the growing reliance on data-based business models, there is now a conflict between commercial goals and individual privacy rights. With social media continuing to develop, it is vital to understand how personal data is collected, stored, and used as well as what this means for both commercial entities and individuals. Balancing new product development with meeting business model objectives while protecting user's personal privacy has emerged as one of the most pressing issues in both academic research and the general population.

2.3 Data Privacy Concepts

The protection of personal information and data refers to the right or power of an individual to choose how his/her data can be gathered, utilized, retained, and provided to anyone else by a third party. Currently, due to the enormous amount of personal information being gathered and shared by different organizations through many means, including social media, Individuals who use the Internet daily are growing increasingly concerned about data privacy. In today's increasingly electronic world, there is an increased need for effective forms of data privacy and protection.

Several principal concepts of data privacy include consent, which defines the requirement for an organization to obtain consent from an individual to collect or use his/her information; transparency, which establishes the requirement for an organization to clearly explain what information will be gathered and how it will be used; and accountability, which establishes the requirement for an organization to manage personal data in an ethical manner and in a secure manner.

Another related principle of data privacy is data minimization which promotes that organizations only gather data that is expressly necessary for a specific purpose, implementing security measures to protect against unauthorized access and/or cyberthreats, and providing individuals with rights to their personal information, including the ability to access, correct, and delete their data when appropriate.

Countries from all over the world have passed new laws/standards regarding their information security procedures, which can include pieces of legislation like the General Data Protection Regulation (GDPR). Additionally, there are many other laws around data protection that have created strong new frameworks for governments around the globe, and companies/institutions will have greater accountability (to users) should they violate a privacy right. This better understanding of privacy helps evaluate the social media platforms and their impact on people.

2.4 Ethical Issues in Social Media

The ways that social media platforms gather and utilize user information raise many ethical concerns. Ethics can be defined as very basic principles that guide our actions fairly and responsibly, particularly when those actions may potentially impact another individual and/or society as a whole. Organizations

that own social media sites are typically more concerned about the profitability, level of engagement, and fulfillment of their business goals than they are about protecting their users' privacy or well-being.

The use of informed consent is one of the largest ethical concerns of using social media. Unless explicitly informed otherwise, users will sign up to use a social media site and agree to the privacy policy, which often includes language allowing the company to access unlimited user information without consent or knowledge of its use. The use of targeted advertising based on user behavior also raises many ethical concerns; particular companies will collect and analyze user information in an attempt to shape users' purchasing behavior, political viewpoint, and/or online activities. According to many critics, these companies use these practices to manipulate their users, and therefore users have lost their personal autonomy.

Additionally, sharing user information with third-parties raises additional ethical concerns, as there are many ways in which user information can be misused, without the consent or knowledge of the individuals whose data is shared.

There is also an ethical issue regarding the obligation of social media businesses to safeguard data from hackers and breaches. Inadequate protection may expose users to identity theft, losses financially, or loss of reputation; because of this, social media companies need to exhibit ethical behavior with regards to transparency, accountability, fairness, and respect for user's rights in their business practices in order to build user trust in technology companies.

2.5 Recent Cases

There have been recent problems related to the collection of social media users' personal data including not protecting user privacy as a potential concern. Therefore, it is important to manage users' personal data in a responsible manner.

One of the more widely publicized events was the Facebook-Cambridge Analytica political scandal in which many millions of users of Facebook had their data used without their consent for purposes of political advertising. The use of personal data without consent raised serious concerns about how that information would impact users' belief in political issues or the outcome of elections. While the investigations about Cambridge Analytica were being carried out by regulators and in the media, Facebook received a tremendous amount of criticism and scrutiny from individuals.

There have been multiple social media sites in addition to Facebook that have had security breaches, with hackers stealing email addresses, phone numbers, and account information from those sites. These breaches have raised concerns regarding the possibility that large technology companies can fall victim to cyber crime/thievery. Users whose data has been exposed as a result of these breaches are at risk of experiencing identity theft and/or fraud, and their accounts being used without authorization.

Many people are expressing their apprehension regarding platforms such as TikTok and the way that they collect data from their users. Various groups that advocate for consumer rights, as well as government entities, have begun to question the manner in which certain countries collect and use their citizens' data and their aged records. Therefore, regulatory reform will result from

ongoing digital sovereignty and international relationships' debates related to national security & data governance.

Current events have shown that there are still privacy risks associated with technology, even with recent advances in digital privacy regulations. They also serve as valuable lessons to researchers, policymakers, & organizations attempting to enhance their data protection policies while operating on social networks.

2.6 Limitations

The limitations of most studies conducted to date regarding social media privacy issues are many. For instance, most research is only done on a certain social media platform or geographic location, which limits the ability of researchers to generalize their findings to different demographics and cultures. In addition, user behavior and privacy expectations are affected by demographic factors, social factors, and technology.

Another significant limitation is that new social media technologies are continuously being developed. As a result, we are left with outdated research results from studies conducted a few years ago, which do not reflect current privacy practices or user experiences. Due to the nature of social media applications, there are no electronic records that can give the researcher a complete understanding of how social media uses information shared by users.

A lot of existing research is based on surveys or self-reported user data. Participants may answer questions differently than intended based on misunderstandings, memory problems, or due to their social stigma associated with providing "incorrect" answers to particular questions. All of these factors contribute to the reliability and validity (or lack thereof) of the research

findings. In addition to these issues, researchers often do not have the ability to gain access to internal company data regarding the operations of social media platforms, limiting their understanding of how a social media platform processes and uses information provided by users.

A number of research studies have looked into the issue of data protection and privacy however; not much has been done on the efficacy of current laws and regulations (for example) or policies by organisations. Also, there is a lack of research into how people are aware of what type of data is being collected from them as well as how to manage their own privacy settings when utilising various websites and applications. These limitations indicate that understanding social media privacy issues remains challenging due to the evolving nature of technology, user behaviour, and data collection practices. It is necessary to identify these research gaps to help direct future researchers to important social media security and privacy issues that could potentially be resolved with appropriate answers.

2.7 Chapter Summary

The chapter provided an overview of the existing literature about social media platforms, practices of collecting personal information, definitions and concepts of privacy, ethical issues associated with these platforms, and recent examples of privacy-related incidents. The literature demonstrates that although the benefits of using social media platforms include facilitating communication and providing access to data, the potential impacts of these platforms raise significant issues regarding privacy, security, disclosure, transparency, and accountability. Prior research has documented many of the difficulties associated with protecting the privacy of users' data in an ever-changing digital world. These findings provide a basis upon which future chapters can perform an analysis of the ethical issues involved with these platforms.

CHAPTER 3 RESEARCH METHODOLOGY

3.1 Introduction

This chapter explains how data privacy issues on social networking sites were investigated in this study through an ethical lens by analyzing how social networks collect, process and use the personal data of their users, the associated risks of violating their privacy with respect to those practices. The importance of having a good research methodology is that it establishes a clear methodical procedure for collecting, analyzing and interpreting information from reputable sources.

In order to address the objectives of the study, which are 1., to investigate data privacy issues that exist in social networks; 2., to analyze data privacy from an ethical standpoint; 3., to identify where and what caused these violations of privacy; 4., to evaluate the effects of these privacy violations on individuals, social networks and society as a whole; and 5., to suggest practical solutions to the problems within the social networking industry regarding the violation of individuals' privacy. In order to satisfy these objectives, a literature-based methodology was utilized in examining secondary sources of data such as research papers, industry reports, government documents and reputable news media.

3.2 Research Approach

This study takes qualitative case study methods into account rather than collecting primary data from study volunteers. Instead, the study is based on literature and documented real living examples concerning data privacy issues on social media platforms.

The use of a qualitative approach is appropriate for this study because it is designed to provide greater understanding of ethical issues, challenges of privacy and how organizations conduct business on social media sites. It allows the researcher to look at existing evidence, note common themes and consider the problem from several points of view.

Case studies will include a review of the Cambridge Analytica situation and the regulatory investigations of TikTok in order to demonstrate the real world examples of the types of privacy challenges social media companies face. In these case studies, there are numerous ethical questions for the company when they collect personal information about an individual, when they obtain consent from users to use that data, when they have transparency (provide individuals with clear notices of how their data will be used) and when they provide accountability (ensure individuals know what their data is being used for), and when they protect the data of the individuals.

3.3 Data Collection

Various reliable secondary sources were used for this study. These reliable sources contained many academic journal articles, conference papers, government reports, regulations, industry reports and news reports from reputable news organisations.

Data was sourced from the Pew Research Centre, Reuters, privacy regulators, academic databases and published social media privacy research concerning the protection of data as they all provide relevant, current, credible and evidence-based information related to this study topic.

The materials collected in this secondary data research were classified by key themes such as: data collection methods; privacy concerns; ethical issues;

regulatory challenges; reasons for privacy violations; the effects of those privacy violations on society.

3.4 Research Activities

Researchers went through multiple steps to investigate data privacy issues related to social media. They first reviewed the literature to gather a better understanding of data privacy and identify key terms, ethical issues, and current events.

After developing a list of key terms, the researchers found and categorized data from several sources based on what objective they were working on in their research. The researchers also analyzed the data they received, which could help them find patterns, similar issues, and significant case studies related to the topic of data privacy.

Researchers have organized and separated the sections of the research report into the following categories: ethical analysis of data privacy issues, root causes of data privacy issues, effects on individuals, and solutions to data privacy issues. In addition to this organizational structure, researchers compared and contrasted information obtained from different sources, confirming that the information obtained is all equal in consistency, reliability, and relevance to their research project objectives.

3.5 Chapter Summary

The methodology employed in this research project is presented in this chapter. A qualitative case study approach using only secondary sources of data was selected to look into issues relating to data privacy on social media

platforms. Secondary sources of data were collected from academic literature, government reports, regulatory findings, industry publications or journals, and trustworthy news sources.

This selected methodology provided a framework for analysing privacy-related issues as well as; ethical issues, causes, effects and possible solutions, associated with social media platforms. The study was able to offer a complete understanding of modern-day data privacy issues within the context of digital practices, by using reliable secondary sources of information and by documenting actual cases in the real world.

The ethical analysis of data privacy issues in social media platforms will be presented in the next chapter based on the data gathered using the methodology presented in this chapter.

CHAPTER 4 ETHICAL ANALYSIS OF DATA PRIVACY ISSUES

4.1 Privacy

One of the primary ethical dilemmas presented by the use of social media platforms is privacy. Today's social media services rely on a vast number of different forms of user data, from standard single source data(i.e., the user's email address) to more complex data types, such as online status, web page history, location tracking, device metadata, online presence and user behavior. Many users do not realize the amount of data their social media accounts are collecting, especially since some of this data is provided voluntarily by the user.

The serious ethical problem is that there is not enough or meaningful informed consent. Most privacy policies are very long and complicated to read, and users will agree to the privacy terms without understanding what they will do with their information. According to figure 4.1, a survey conducted by the Pew Research Center, 56% of respondents said they frequently accepted the privacy policies without reading them , and 69% said they saw these policies as just another thing that had to be cleared before moving forward. [1]

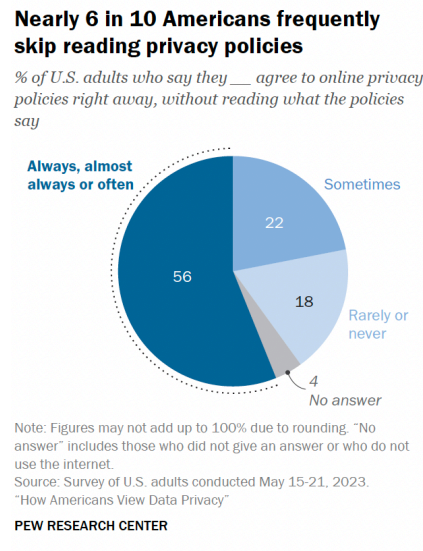


Figure 4.1 Skipped reading privacy policies

Source: Pew Research Center (2023), How Americans View Data Privacy.[1]

The principle of privacy requires that organisations respect and individual's right to control their personal information. Many social media companies use this collection of data as part of their business model, raising questions as to whether or not users have control over their own data

A prime example of such exploitation of user data can be observed in the Cambridge Analytica case. There was a breach of trust because Facebook allowed Cambridge Analytica to access to the data of approximately 87 million users without acquiring explicit consent. What this incident illustrated is that user data is not only available for its intended purpose but may be abused. [2]

Thus, the ethics of social networks hinges upon our ability as organizations to demonstrate transparency, to get informed consent, and to take appropriate precautions to protect users' data. Not doing so can lead to a loss of

confidence in service providers and result in the large-scale use of personal information without user consent.

Privacy Concern	Description	Key Finding / Example
Lack of Informed Consent	Users agree to privacy policies without understanding how their data will be used.	56% of users frequently accept policies without reading them (Pew Research Center, 2023).
Data Exploitation	User data is used beyond its original, intended purpose, often as a core business model.	The Cambridge Analytica case involved access to ~87 million Facebook users' data without explicit consent.
Consequences of Failure	Lack of transparency and consent leads to a loss of user trust and potential misuse of personal information.	Service providers face a loss of confidence and large-scale unauthorized use of personal data.

Table 4.1: Summary of Key Privacy Concerns and User Consent Issues on Social Media Platforms

4.2 Security

Security encompasses the safeguarding of private data from unauthorized access, misuse and/or disclosure. Many of the sites and applications used for social media result in high volumes of private data being stored within them, creating many reasons for cybercriminals to target these platforms.

Sensitive user information that is available due to a data breach includes the user's email addresses, passwords, credit card details and social security numbers. Failure to secure user data can result in identity theft, financial fraud, and unauthorized access to user accounts.

Pew Research Center says that 34% of American adults have encountered at least one cyber-related event (e.g., identity theft/fraud) within the last year (Figure 4.2). This statistic indicates the high level of risk that people face regarding cybersecurity when using digital devices. [3]

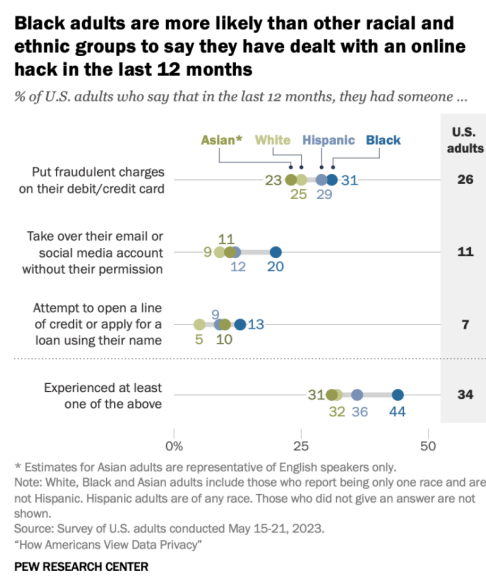


Figure 4.2 Experienced at least one cyber-related event

Source: Pew Research Center (2023), Key Findings About Americans and Data Privacy.[3]

The incidents involving Cambridge Analytica have also indicated that Facebook is insufficiently monitoring third-party access to the user data collected by their platform. A significant amount of user data can be collected from Facebook's platform due to its design, creating significant and substantial security risks for users. [2]

Moreover, Canadian privacy regulators have raised concerns about the security of TikTok based on an investigation into how the company stores and uses children's personal information. One finding of this investigation is that TikTok's methods for preventing minors from accessing the platform were inadequate, and thousands of children had their biometric (e.g., voice and facial recognition) and location data illegally collected. As a direct result of its investigation, the regulators determined that TikTok collected a significant amount of sensitive data related to children, including biometric (i.e., face recognition, voice recognition) [4]. When considering possible impacts from a breach or unauthorized access or misuse of personal information, the sheer volume of sensitive information kept by the company increases the risk an incident would have on users and their families. The report also indicates that issues may exist regarding cross-border transfer of children's personal information to jurisdictions outside of Canada; as well as, general access/accessibility of user information (e.g., children's) stored on systems in jurisdictions where it was not collected.

The findings of the investigation indicate that strong security controls, restricted access, and protections for data or sensitive personal information are required anytime an organisation processes sensitive information belonging to its users. Failure to implement necessary security will expose individuals who access the service to significant privacy and security risks, particularly if the group accessing the service is at risk (e.g., children). Additionally, social

media services truly have an ethical responsibility to protect sensitive personal information from being accessed by individuals; and/or misused.

From an ethical perspective, social media organizations are ethically responsible for putting strong cybersecurity procedures in place in order to protect customer data. Organizations must supply adequate security controls to meet this ethical duty based on the principle of security, through providing security assessments routinely, and by responding to new threats promptly. When organizations do not meet these requirements, cybercrime risk is increased and public trust in the use of social media technologies is eroded. Consequently, organizations must maintain adequate security protocols in order to sustain users' confidence in social media technologies and to ethically use social media technologies.

Security Threat	Description	Example / Finding
Data Breaches	Unauthorized access to sensitive user data stored on social media platforms.	Exposed data can include email addresses, passwords, credit card details, and SSNs.
Insufficient Third-Party Monitoring	Platforms fail to adequately control how third-party apps access and handle user data.	The Cambridge Analytica incident highlighted Facebook's failure to monitor third-party access.
Vulnerability of Sensitive Groups	Specific platforms lack adequate security for vulnerable users, increasing their risk.	Canadian regulators found TikTok inadequately protected children's data, illegally collecting biometric and location data.

Potential Impact on Users	The direct harm resulting from security failures.	34% of American adults experienced at least one cyber-related event (e.g., identity theft) in the past year (Pew Research Center, 2023).
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Table 4.2: Security Threats, Platform Failures, and Associated Risks to Users

4.3 Justice

When speaking about the concept of justice, this can refer to fairness and/or equality in the gathering, processing, and use of an individual's personal data. As it relates to social media platforms, the concept of justice requires that all users are treated fairly, and that their personal information is used in ways that do not create discrimination, manipulation, or unequal results. The ethical issues associated with using algorithms and other data-driven systems in order to provide personalized content, advertisements, and recommendations raise the potential for unintended consequences to occur by either favoring certain groups while disadvantaging others.

Many users provide personal information to social media platforms with an incomplete understanding of how this information is likely to be used. The data collected from users can also be used for targeted advertisements, behavior-based profiling, and political campaigning, among other applications. A prime example of the unethical use of personal data to influence public opinion and voting behavior was the Cambridge Analytica scandal, where the personal information of millions of Facebook users was utilized to create psychological profiles and deliver targeted political messages.[2] This event demonstrated the types of injustices that can arise when personal data is misused to manipulate individuals into voting for or against a specific candidate, thereby creating significant questions about both the fairness and

ethicality of using individuals' personal information without their consent. By not giving users equal knowledge or control with respect to how their personal information was being utilized, these practices create considerable challenges regarding their compliance with the principles of justice.

All users must obtain equal protection through justice regardless of characteristics such as age, nationality, gender, etc. Evidence from investigations into TikTok's data collection show that children are at higher risk of having their sensitive personal information collected and processed without appropriate protections.[4] Children might not completely understand what the implications of data collection are, making them especially vulnerable to potential exploitation and privacy risks. This proves that there is inequity in protection for vulnerable populations; namely, children are at greater risk than all other users.

In addition, algorithm-based systems on social media platforms contribute to inequity through how algorithms determine how users will experience which content and what opportunities they will be given. Personalized recommendation systems can perpetuate existing bias, result in information bubbles, and provide users with different experiences based only on their demographics or website activities; as illustrated in Figure 4.3, algorithmic decision-making can affect content and ad visibility for users, resulting in unequal access to content and ads by users.

Organizations have an ethical obligation to create and maintain systems based on data meaningfully within their parameters. In addition to having meaningful systems, social media sites must regularly audit their algorithms for possible biases, to mitigate any possible means of manipulating how people access information, how people are treated when using information, and/or providing equal access and protection to everyone in society. If these systems are designed, created, and/or operated without a consideration of

ethical principles of fairness and justice, it is possible for manipulation to occur, unequal access to information to exist, as well as unduly different treatment of people based on their ethnicity/race, age, or other factors violating ethical principles of fairness and justice. Table 4.3 summarizes several justice-related concerns associated with the collection and use of personal data on social media platforms.

Justice Concern	Description	Example
Data Exploitation	Personal data used beyond user expectations	Cambridge Analytica political profiling
Unequal Protection	Vulnerable groups receive insufficient safeguards	TikTok children's data collection
Algorithmic Bias	Algorithms may favor certain groups or viewpoints	Personalized content recommendations
Information Manipulation	Users influenced through targeted content	Political advertising based on behavioural data

Table 4.3 Ethical Justice Concerns in Social Media Platforms

4.4 Accountability

Accountability is one of the fundamental ethical principles related to safeguarding data privacy. It defines who will be held accountable when personal data has been misused, violated, or collected without proper consent. Given how social media companies collect an extremely large volume of

personal data, determine how it is utilized, and distribute the data to a wide range of third parties, the issue of determination of accountability is critical in instances when individuals' privacy is violated.

When privacy violations occur in the realm of social media there are frequently questions as to if the responsibility lies with the provider, third-party software developers, regulatory authorities, or the user themselves.

One of the primary issues of concern regarding accountability in terms of social media-related issues is a lack of transparency with respect to how social media companies collect and use information. Many social media companies argue that users have consented to the collection and use of their personal information by accepting the terms and conditions of the social media platform. However, many users believe that they have not provided informed consent as to how their information will be collected, analyzed, used, or monetized. Therefore, while technically valid in law, the concept of consent may not necessarily be ethically valid.

The decrease of public belief in social media companies being accountable has been dramatic. Pew Research Center released a report stating that out of the American respondents surveyed, 77% indicated no or little trust that executives of social media companies would be honest about wrongdoing, and accept responsibility for any wrongdoing concerning user data. Of those participants, 71% had no or little confidence in the government having the ability to compel social media companies to become accountable for violations of user privacy [3]. As evident from Figure 4.4, the overwhelming majority of respondents did not feel confident that social media companies' managers will accept responsibility for the misuse or wrongful use of data, nor that any government can enforce regulations against such an entity. The results of this

survey indicate that a majority of users believe that the current methods of accountability are inadequate to protect their personal information.

Most Americans don't trust social media CEOs to handle users' data responsibly

*% of U.S. adults who say they have **very little or no trust at all** that leaders of social media companies will ...*

Publicly admit mistakes and take responsibility when they misuse or compromise users' personal data **77%**

Not sell users' personal data to others without their consent **76%**

Be held accountable by the government if they misuse or compromise users' personal data **71%**

Note: Those who did not give an answer or who gave other responses are not shown.

Source: Survey of U.S. adults conducted May 15-21, 2023.
"How Americans View Data Privacy"

PEW RESEARCH CENTER

Figure 4.4 Most Americans Do Not Trust Social Media CEOs to Handle Users' Data Responsibly

Source: Pew Research Center (2023), Key Findings About Americans and Data Privacy.[3]

The Cambridge Analytica scandal is perhaps the clearest example of the failures of accountability in the social media industry. This was when it was revealed to the public, and to many regulators around the world, that Facebook permitted a third party application to collect personal data from approximately

87 million users for the purpose of using that personal data for political profiling, without obtaining explicit consent. After the public outcry over this incident, Meta (formerly Facebook) has faced numerous investigations, lawsuits, and regulatory actions in several countries throughout the world [2]. The Cambridge Analytica case is an example of the negative consequences resulting from a lack of oversight of social media companies, and how a company must be held accountable for their own actions, and for the actions of third-party organizations that are allowed access to their users' data.

From the ethical standpoint, Organizations should have transparency, have proper governance and controls in place, comply with any applicable regulations and provide their users with notice and access to complain and seek a remedy for any harm suffered. If an organization fails to meet its obligations through the aforementioned actions, it can result in privacy violations, liability, damage to its reputation and a loss of public trust. As a result of these issues, accountability is an underlying ethical requirement for responsible data management practices related to social media platforms.

Accountability Issue	Description	Supporting Evidence / Example
Blame Diffusion	It is unclear who is responsible when a violation occurs (provider, third-party, regulator, or user).	Social media companies cite "consent" via T&Cs, while users argue consent was not informed.
Lack of Public Trust	Users do not believe that companies or regulators will enforce accountability.	Pew Research Center found that 77% have little or no trust in social media CEOs to accept responsibility for wrongdoing.
Regulatory Weakness	Belief that government lacks the ability to compel companies to be accountable.	71% of respondents have little or no confidence in the government's ability to enforce accountability (Pew Research Center, 2023).
Failure of Oversight	A clear example where a lack of accountability led to massive harm.	The Cambridge Analytica scandal, where Facebook allowed third-party access to 87 million users' data for political profiling without explicit consent.

Table 4.4: Accountability Challenges, Public Trust Deficits, and Oversight Failures

CHAPTER 5 REAL CAUSES AND IMPACTS

5.1 Real Causes of the Problem

5.1.1 Data-Driven Business Models

Generally speaking, most social media companies focus on advertising for revenue generation rather than direct subscription fees to users. Thus, advertising is the primary source of company revenue for social media platforms, and the way they achieve this is by selling advertising to businesses. In addition, social media companies have access to user data that can be analysed to create more target-specific campaigns. Therefore, since social media companies possess such a considerable volume of valuable user data, it allows companies an advantage in optimising their advertising space based on user data.

Because advertisers use social media sites for advertising purposes, social media sites currently track and analyse extensive amounts of user-specific data, including consumer behaviour records, past purchase history, and location-based interests. Researchers have examined how many pieces of data advertising companies collect and what they do to more effectively use this data for targeted advertising. As a result of this information, companies can create extremely detailed user profiles that allow them to best market to consumers by predicting what products or services a consumer will purchase next. [5]

Moreover, social media companies have tremendous financial incentives to continue expanding their data collection methods and creating innovative ways to acquire the relevant information about the user.

The modern complexity of digital advertising systems shows how users' personal data moves through a diverse array of entities before being delivered to them with their ads. (Figure 5.1.1) illustrates the mobile ad ecosystem, in addition to the interaction between users, advertisers, analytics providers, and advertising platforms.

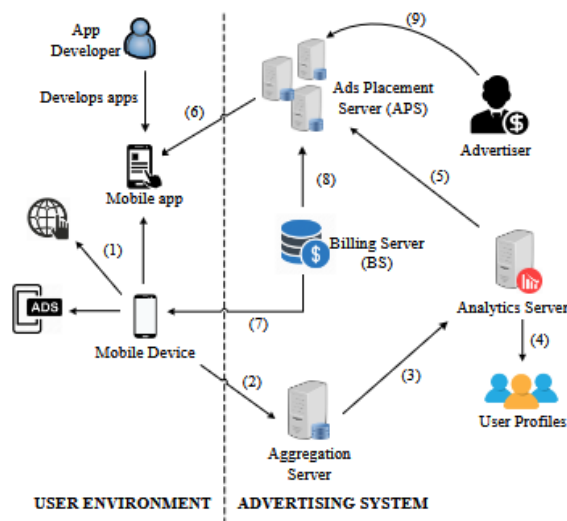


Figure 5.1.1

The Mobile Advertising Ecosystem and Information Flow Among Different Parties

Source: Ullah, I., Boreli, R., & Kanhere, S. (2020) [5]

As presented in figure 5.1.1, user data goes through a multitude of stages of collecting, aggregating, profiling, and delivering ads. This flow of data gives too many different parties potential access to the same personal data; therefore raising issues relating to privacy protection, user consent, and accountability.

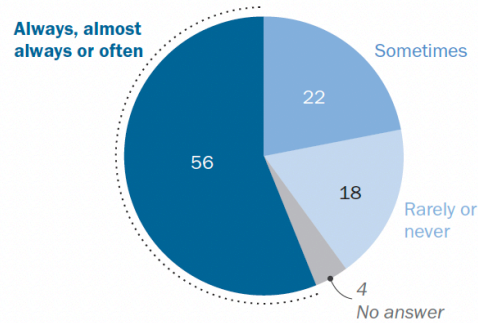
This business model creates many data privacy problems for those who use social media. Because there is an incentive for a company to maximize advertising revenue, they may collect an unnecessary amount of personal information. This results in a higher risk of being affected by privacy violations, sharing of personal information without permission, and using someone's personal information in ways not originally intended. Thus, the ongoing struggle between the commercialization of social media and protecting user privacy represents one of the most critical ethical dilemmas in operating a modern social media platform.

5.1.2 Lack of User Awareness

Many users do not really know about how their data is collected, processed, and shared or even whether they have control over their privacy settings. In addition, the company's policy on how they track users and the resulting impact of the data they collect on the user is often not understood. According to the Pew Research Centre, a recent study showed that the majority of users will typically accept the privacy policy without having read it [1]; see Figure 4.1. This shows a general lack of engagement by the average user with the terms and conditions regarding data protection set by the companies they are dealing with.

Nearly 6 in 10 Americans frequently skip reading privacy policies

% of U.S. adults who say they ___ agree to online privacy policies right away, without reading what the policies say



Note: Figures may not add up to 100% due to rounding. "No answer" includes those who did not give an answer or who do not use the internet.

Source: Survey of U.S. adults conducted May 15-21, 2023.
"How Americans View Data Privacy"

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Figure 4.1 Skipped reading privacy policies

Source: Pew Research Center (2023), How Americans View Data Privacy.[1]

Users' lack of knowledge about their data collection abilities severely limits their ability to make educated choices regarding their data. Because of this, in many cases, users are giving their consent to have their data collected, used and/or distributed, without understanding what they are consenting to. This raises ethical concerns surrounding informed consent because users do not have any meaningful way to control how their data is used or what kinds of consent they are providing to the companies that collect their data. Because users are not informed about their data collection practices, the information imbalance between users and the companies collecting their data creates an ongoing risk to user privacy in the digital world.

5.1.3 Weak Regulatory Enforcement

Although laws regulating data protection worldwide are in place to protect users' data, such as The General Data Protection Regulation (GDPR) for Europe and various national regulations, there are still many challenges facing regulators when trying to effectively enforce data protection regulations. One of the main challenges facing regulators when trying to enforce compliance with privacy laws is that social media platforms operate across multiple jurisdictions and have many different laws and regulations in each of those jurisdictions that they have to comply with in addition to the fact that each of those jurisdictions have different levels of enforcement ability. This can make it difficult to ensure that social media platforms are complying with rules regarding privacy protection consistently across the globe.

In addition, large technology companies also have lots of resources to deal with these types of issues. They have lots of financial, legal, and technical resources that they are able to utilize to complicate the enforcement process for regulators. Regulators may not have the ability to monitor companies' data practices in real time, which also makes it hard to enforce penalties against companies that violate privacy regulations.

The case of TikTok is an example of this challenge. In Canada and numerous other countries, there have been multiple investigations by various national and international regulators into TikTok's data handling practices involving transparency, consent, and cross-border data transferring [6]. These investigations demonstrate the continued difficulty that regulators are having in forcing compliance through regulatory enforcement of privacy laws with fast-moving businesses utilizing global virtual data storage.

Regarding the ethical implications of weak enforcement of privacy laws, this erodes the principle of accountability. Organizations may not be held accountable for violating these laws in a timely or effective manner, thus creating a disparity between the actual legal framework and enforcement within society. This further increases the potential for continual abuse of personal information via social media.

5.1.4 Rapid Technological Development

Digital technology has advanced so much that the legislation and regulation to support it have not been developed quickly enough. New technologies and tools such as AI, recommender systems based on machine learning, and Big Data allow companies to collect, process and analyse large amounts of consumer information. These technologies are capable of processing large quantities of data in very short time frames, which is too fast for current regulatory regimes to effectively monitor and may create potential ethical issues with regard to using of data appropriately and in a compliant manner.

As a result, technology is outpacing the laws and regulations that are currently in place. The fact that there are either no regulations or very little regulation in terms of the collection and usage of information obtained from consumers creates many market opportunities for companies to collect data regarding their customers without providing any form of restriction on how their collected data is going to be used (or for that matter ethical restrictions regarding how the data collected will be used). This results in the potential for increased unethical/unjustified profiling of consumers and provides no visibility to consumers on how they provide their personal information to a third party (e.g., a third party who buys that person's profile).

According to a Pew Research Center study (2023), there is a significant body of evidence indicating a lack of confidence among consumers regarding the adequacy of existing laws to adequately protect their personal data. For example, according to this study, it is very clear that many users simply do not know the identity of the regulatory source and/or authority regarding data collected from them and how that data will ultimately be used. This lack of understanding is even more problematic when you consider the rapid rate of technological change occurring today and how that impacts consumers' lack of understanding on this issue. [7]

Moreover, in this analysis it was determined that 72% of Americans don't know anything about data privacy laws with only 27% claiming that they have a solid understanding of them. The low level of comprehension and understanding of data privacy laws has created a perception in the general public that there is little to no control over how their personal information will be collected, stored or utilized by corporations and the government [7]. This is visually depicted in Figure 5.1.4

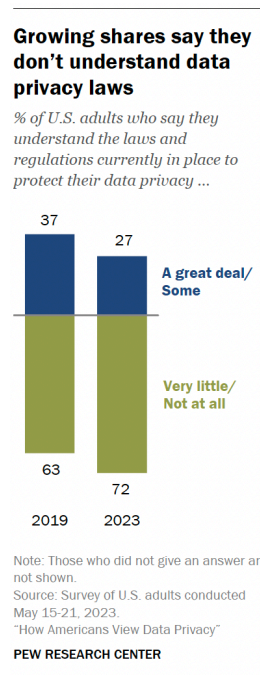


Figure 5.1.4

Growing shares say they don't understand data privacy laws

Source: Pew Research Center (2023) - Views of data privacy risks, personal data and digital privacy laws

The results from this research demonstrate that there continues to be an absence of public awareness about the right to privacy that is given by law, in the U.S. As technology continues to progress there will be an even bigger disparity between the law and the public's understanding and awareness thus creating significant ethical problems with regard to issues like transparency, consumer education, and whether the current data protections provide enough security for people's right to privacy.

The study further supports the ethical argument that regulatory systems in place today are not keeping up with the fast-paced development of new technologies. Therefore, the widening gap between the fast-paced creation of new and advanced technologies, and the slow progression of regulatory systems responding to creating policies for those technologies is one of the major contributors to creating significant problems regarding consumer privacy on social networking sites. Consequently, as long as legal systems remain unchanged and incapable of providing enforcement for their policies, the risk of ethical violations would continue to increase as the pace at which new technology is introduced to society continues.

Cause	Description	Ethical Issue
Data-Driven Business Models	Advertising incentives encourage extensive data collection	Privacy, Consent
Lack of User Awareness	Users accept policies without understanding them	Informed Consent
Weak Regulatory Enforcement	Difficulties enforcing privacy laws globally	Accountability
Rapid Technological Development	Technology evolves faster than regulations	Transparency, Fairness

Table 5.1: Major Causes of Data Privacy Problems in Social Media Platforms

5.2 Impact on Individuals

Violations of privacy within social media can cause various serious outcomes for users. Identity theft, financial fraud, anxiety and stress, loss of personal freedom and reduction in trustworthy use of online services are some of the

expected consequences from the exposure and misuse of personal data by others. It is common for individuals to experience both emotional and financial harm from privacy violations, especially when it involves personally identifiable information, such as their username and password, financial information and personal communication.

A sizeable number of users have suffered from experiencing fraud, account hacks and/or wrongful use of their personal information due to security or privacy failures associated with digital platforms. The Pew Research Center (2023) indicates that many people--in some cases most--addressed cyber issues related to information being exposed due to a lack of adequate protections against privacy violations in an online environment. [3]

The findings that are associated with the subject matter indicate that the violations of privacy do make a difference for real people, and therefore emphasize that there is a need for ethical, as well as technical, safeguards to be built into the design and operation of social media systems.

Type of Impact	Description	Example / Finding
Financial Harm	Identity theft, financial fraud, unauthorised use of personal information.	Exposure of usernames, passwords, financial data.
Emotional & Psychological Harm	Anxiety, stress, loss of personal freedom.	Consequences from misuse of private communications.
Prevalence	Many users have experienced fraud, account hacks, or wrongful data use.	Pew Research Center (2023) documents widespread cyber-related incidents [3].

Overall Conclusion	Privacy violations cause real, tangible harm to individuals.	Ethical and technical safeguards are necessary.
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Table 5.2: Direct Harms to Individuals from Social Media Privacy Violations

5.3 Impact on Organizations

There are many repercussions for organizations that operate social media platforms when they experience breaches in their data privacy and security. These will include being fined by regulators, having users or the regulatory bodies file lawsuits against them, suffering tremendous damage to their reputation, losing users' trust, and losing market capitalization. In an aggressive digital marketplace, as is the case in today's business environment, just one major privacy breach can result in long-lasting damage to the financial and reputational standing of an organization.

Cambridge Analytica was a big scandal that brought a lot of backlash to Facebook following an inquiry into how Facebook used people's data without permission or transparency. Cambridge Analytica raised many issues regarding Facebook's collection, sharing, and monetizing personal information of users. The week after the incident (known by The Guardian as a week of shame for Facebook) saw Facebook receiving lots of negative media coverage and widespread distrust by users of how the company managed their data. [8]

Both ethical and commercial considerations come into play when assessing the effects of events like Cambridge Analytica on global society. The immediate consequences of an organization's inability to protect data from misuse will be

the likelihood of damage and will expose the organization to additional risk (by way of potentially experiencing a legal claim).

Legal risks may entail potentially large monetary sanctions; however, sanctions could be more than simply financial penalties; they may continue for long periods of time to erode an organization's reputation, marketing ability, and ability to attract new customers. To be successful in the digital world, organizations must provide ethical oversight and create strong privacy protection processes so that their presence in the digital marketplace can continue to be viable, credible, trustworthy, and sustainable.

Consequence	Description	Example
Regulatory & Legal Penalties	Fines, lawsuits from users or regulators.	Potential large monetary sanctions.
Reputational Damage	Loss of user trust, negative media coverage, reduced market capitalisation.	Cambridge Analytica scandal – “week of shame” for Facebook [8].
Operational & Marketing Harm	Long-term erosion of ability to attract new customers.	Distrust persists after major breaches.
Ethical & Commercial Necessity	Organisations must implement ethical oversight and strong privacy protections.	Needed for credibility, trust, and sustainability.

Table 5.3: Organizational Consequences of Privacy and Security Failures

5.4 Impact on Society

Privacy violations on social media can have consequences that go beyond the individual level and organisations to the larger society. An example of this is the rise of digital surveillance as large quantities of data can be collected and analysed to build user profiles in a continuous manner allowing for constant tracking of their behaviours online (which ultimately reduces their level of privacy in these digital settings). As a result, this raises concern that these types of continuous monitoring will become normalised within society as a whole.

Targeted delivery of information (individualised feeds based on an individual's behaviour) means that privacy violations will alter how the public views things because they are influencing people's decision-making processes and limiting their exposure to differing opinions.

Another social effect of privacy violations is that they can diminish the amount of trust the public has in companies or platforms engaged in collecting, analysing or possibly misusing their personal data. The erosion of public trust can also extend to other data driven systems and institutions in society that are reliant on it for decision making. Privacy violations can also create dilemmas for democratic systems because they allow for ways in which to use personal data in political advertisements, conduct behavioural targeting and utilise misinformation against voters.

In general, these problems show the public's increasing worry about the administration of data and the responsibilities businesses must accept when using your information. The Pew Research Centre reports that many people are worried about how companies are collecting and using their personal data (Figure 5.4) , and a lot of people have little to no confidence that social media

companies are using your data in a responsible way (Figure 4.4) [1]. This is a sign of a major increase in people realising the privacy risks that exist; thus, they need better regulations, more transparency and accountability from businesses in relation to their usage of user data in the digital arena.

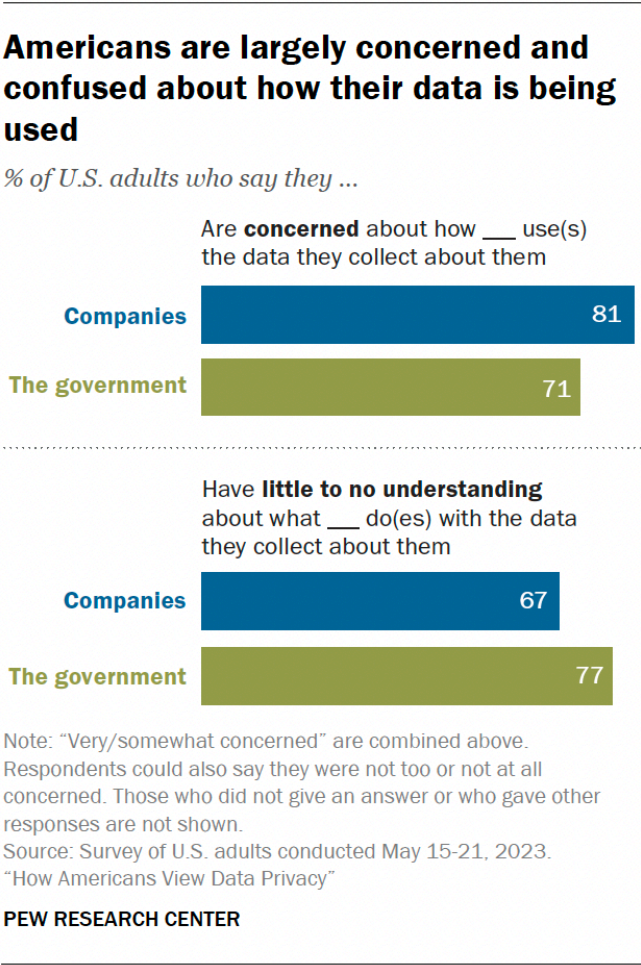


Figure 5.4 Americans are largely concerned and confused about how their data is being used

Source: Pew Research Center (2023), *How Americans View Data Privacy*. [1]

Most Americans don't trust social media CEOs to handle users' data responsibly

*% of U.S. adults who say they have **very little or no trust at all** that leaders of social media companies will ...*

Publicly admit mistakes and take responsibility when they misuse or compromise users' personal data **77%**

Not sell users' personal data to others without their consent **76%**

Be held accountable by the government if they misuse or compromise users' personal data **71%**

Note: Those who did not give an answer or who gave other responses are not shown.

Source: Survey of U.S. adults conducted May 15-21, 2023. "How Americans View Data Privacy"

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Figure 4.4 Most Americans Do Not Trust Social Media CEOs to Handle Users' Data Responsibly

Source: Pew Research Center (2023), Key Findings About Americans and Data Privacy.[3]

Societal Impact	Description	Supporting Evidence / Example
Normalisation of Digital Surveillance	Continuous tracking of online behaviour becomes accepted.	Reduction of privacy in digital settings.
Manipulation of Public Opinion	Targeted content feeds limit exposure to differing views, influence decision-making.	Use of personal data in political advertising and behavioural targeting.
Erosion of Public Trust	Distrust extends beyond individual companies to other data-driven systems and institutions.	Pew Research: many people are concerned about data collection (Figure 5.4) and have little confidence in social media CEOs (Figure 4.4) [1][3].
Threats to Democracy	Misinformation, voter profiling, and behavioural manipulation undermine democratic processes.	Increased public demand for better regulations, transparency, and accountability.

Table 5.4: Broader Societal Harms Arising from Privacy Violations on Social Media

CHAPTER 6 PROPOSED SOLUTIONS

6.1 Technology Solutions

To decrease data privacy threats on social media, technology solutions are vital resources. Social media providers collect a huge volume of user information (e.g., emails, profiles, photos), process it, and store it without sufficient safeguards to stop unauthorized access or protect against misuse or abuse of that data from cybercrime.

A key technology solution would be multi-factor authentication. Multi-factor authentication would provide users with another method of authenticating besides providing only a password; that is, the user and someone who has obtained a user's password can no longer be considered equal partners when using that password. Multi-factor authentication has gained traction because of the many ways an individual can be authenticated (e.g., via SMS or voice, or by using biometric means).

As a result, social media companies should promote or mandate that users enable multi-factor authentication on all of their accounts containing private or sensitive personal data.

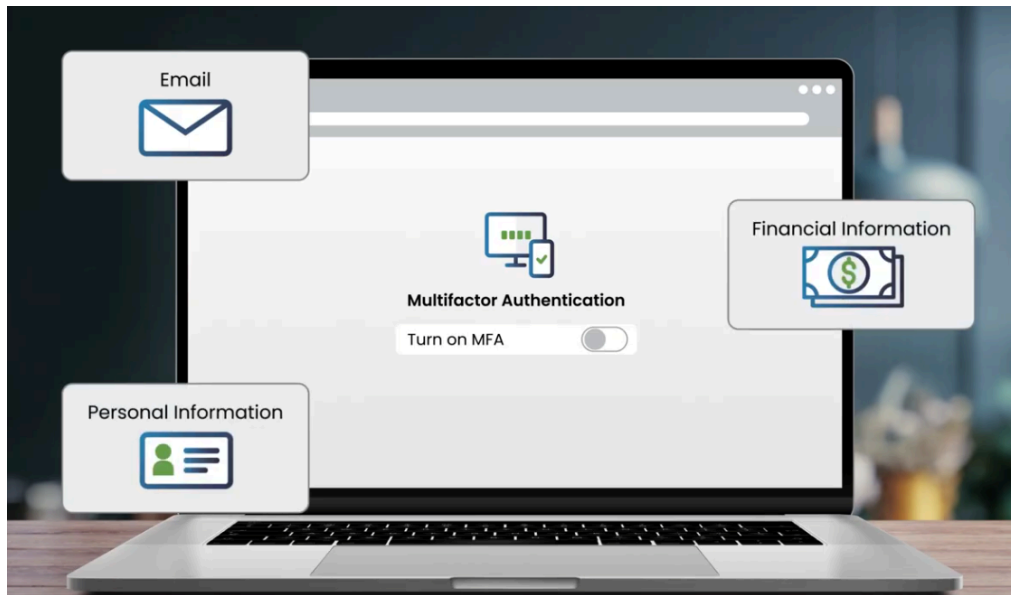


Figure 6.1 Multifactor Authentication as a Security Solution

Source: Cybersecurity and Infrastructure Security Agency (CISA),
Multifactor Authentication. [9]

A key solution is encryption from end to end. Through the use of encryption, private communications and private messages are kept secure until one of the users chooses to render them insecure (for instance, by forwarding the message to someone else). If a message is intercepted by an attacker, they will not be able to read or use the encrypted data, unless they have the key to decrypt it. The encryption can be applied to messages sent on social media or other private mediums, as well as stored personal data and sensitive information (for example, email addresses, phone numbers, passwords, and location information).

Social media platforms should also implement systems based on the principle of privacy by design. Privacy by design means that systems are developed with privacy protections built in from the outset rather than implemented after problems arise. For example, social media platforms should only collect the personal data required to provide the service, offer clear privacy settings, and

provide end-users the ability to control how their personal data is used by the platform. This will both reduce the amount of unnecessary data collected by social media platforms and increase user trust in the platforms.



Figure 6.2 Principles of Data Protection

Source: European Data Protection Board (EDPB), Principles of Data Protection. [10]

Another significant thing to keep in mind is the use of data minimization as part of a company's technology solutions for improving social media data privacy. When social media companies use as little personal information as possible, there is less risk that people will lose their privacy rights when a data breach occurs. Therefore, businesses must also make sure that they destroy any old or unused personal information as soon as they are no longer needed. By limiting the number of personal records being exposed in a cyberattack, they can help mitigate any potential risks associated with cybercrime.

Ultimately, using technology solutions such as multi-factor authentication, encryption, privacy-by-design and data minimization will help reduce the data privacy risks encountered by social media users. Nevertheless, relying solely on technology will not be adequate; policies, user education and organizational accountability must be present as well.

6.2 Policy Solutions

Policy solutions are required as technology by itself cannot adequately address data privacy issues. Greater regulation is needed from governments/regulators to make certain that social media companies behave responsibly with regard to people's personal information. There should be stronger privacy legislation, which requires that social media companies develop specific written policies describing:

What they are collecting;

Why they are collecting it;

How long they plan to retain it;

Whether they will share with third parties.

Providing this information will assist users in understanding their rights and making more informed choices prior to consenting to the privacy terms they have agreed to.

In conjunction with providing clear written data policies; greater penalties should exist for companies who misuse user data or do not take adequate steps to secure an individual's private data. Many large corporations may not place a premium on privacy protections if there is too little risk of being penalized for those failures. Increased penalties may induce organizations to enhance their data security and privacy practices.

Regular audits of social media companies should also be required. Independent auditors should evaluate social media companies' adherence to applicable laws governing the collection and use of personal information, as well as ethical business practices. These audits can detect data handling, third-party data sharing, and data security control weaknesses before there is a significant privacy violation.

Policymakers also need to require that companies create privacy policies that are more easily understood. Because most users do not read their company's privacy policies because they are too long and complex, companies should also establish simple summaries of their respective privacy policies—especially in relation to important issues such as data collection, third-party information sharing, targeted advertising, and how users can delete their accounts. Such actions will enhance the user's ability to provide informed consent and will help minimize the chance of misunderstanding between users and companies.

6.3 User Responsibilities

Users are responsible for safeguarding their own personal data through the use of social media. While businesses have a responsibility to protect the data of their users, it is equally important for individuals to exercise caution when disclosing or revealing information about themselves online.

A first step taken in safeguarding your data is to regularly check your privacy and security settings. Many social networks allow users to choose who can access their usernames, photos, locations and/or contact information (if applicable) or a combination of these profile features. By periodically reviewing these privacy/security features, users will have less chance of accidentally exposing their private information to strangers or third parties.

A second step to take in protecting your identity is reducing the number of personal items an individual shares publicly. Data such as home address(es), phone numbers, school information, financial information, and personal documents should not be disclosed in regular terms to the general public. The act of oversharing these types of items increases the chances of someone stealing another person's identity, doing a fraudulent transaction using a victim's name, threatening the victim; and/or causing serious harm to an innocent person.

A third and very important step to take in protecting your personal identification is being knowledgeable about online scams and fraudulent messages. Many cyber criminals use entity impersonation, unsolicited emails that appear to include legitimate websites, and/or fake log in pages to unlawfully obtain private information from the victim. Whether an entity such as Facebook, Twitter, and/or other social networking sites is sending you a request for your username, password, or any other document for verification, do not provide them.

In conclusion, individuals need to make an effort to know the privacy policies they agree to before using a social media platform. Even if the privacy policy is lengthy, users should be familiar with key components of the policy, including what information is collected, whether their information will be shared with any third parties, and each platform's ability for users to control who can access their information. These efforts will give users greater awareness of their online privacy, allowing them to use social media more securely.

6.4 Organizational Responsibilities

Social Media Companies: Responsibility for collecting, storing, processing, and sharing personal information lies with social media companies. Therefore, these companies should responsibly collect, process and use their members' personal data with integrity.

1. Organizations must be transparent about their use of personal data. Organizations must disclose how they will use personal data after it has been collected, and they should not include important information in long, technical privacy policies that are difficult to read and therefore make it almost impossible for users to provide valid consent.
2. Organizations must not use their users' PII for other than the originally intended purpose of collecting PII, without the express permission from the user. Companies cannot target users for transient purposes, use PII to unfairly classify them or disclose PII to a third party without informing the user of the disclosure. The responsible stewardship of personal data is important to honouring the rights of individuals and establishing public trust in organizations that collect PII.

3. Organizations must invest significant resources in information security. Organizations must establish strong security protocols; conduct regular assessments of their security mechanisms; provide adequate employee training regarding best practices for security of information; and establish adequate procedures for responding to incidents of information security breaches. Moreover, organizations must notify parties affected by data breaches in a timely manner and take appropriate steps to mitigate any harm that results from the data breach.

A sound data security plan is built on 5 key principles:

1. [TAKE STOCK](#). Know what personal information you have in your files and on your computers.
2. [SCALE DOWN](#). Keep only what you need for your business.
3. [LOCK IT](#). Protect the information that you keep.
4. [PITCH IT](#). Properly dispose of what you no longer need.
5. [PLAN AHEAD](#). Create a plan to respond to security incidents.

Figure 6.3 FTC Principles for Protecting Personal Information

Source: Federal Trade Commission (FTC), Protecting Personal Information: A Guide for Business. [11]

In conclusion, organizations must be held responsible for their own actions as well as the actions of third-party organizations that gain access to user information. Companies offering third-party apps or advertisers access to personal information must adequately monitor those organizations. These measures can prevent instances of the collection or misuse of user information without proper user consent.

Overall, the proposed solutions demonstrate that preserving the privacy of data will require cooperation between the technology industry, governmental

policy, users and organizations. If all parties take their responsibilities seriously, social media platforms will become safer and more ethical for users.

Solution Category	Proposed Solution	Description	Related Ethical Issue
Technology Solutions	Multi-factor authentication	Requires users to verify their identity using another method besides password, such as verification code, biometric login or authentication app.	Security
Technology Solutions	Encryption	Protects private messages and sensitive data from being read or misused by unauthorized parties.	Privacy and Security
Technology Solutions	Privacy-by-design	Builds privacy protection into the system from the beginning of development.	Privacy and Accountability
Technology Solutions	Data minimization	Reduces unnecessary data collection and keeps only data that is needed for a clear purpose.	Privacy

Policy Solutions	Stronger privacy laws	Requires platforms to clearly explain data collection, storage, third-party sharing and user rights.	Accountability
Policy Solutions	Regular audits	Checks whether social media platforms follow data protection laws and ethical standards.	Accountability and Security
User Responsibilities	Review privacy settings	Helps users control who can see their profile, posts, location and personal information.	Privacy
User Responsibilities	Limit personal sharing	Reduces risks such as identity theft, scams and cyberstalking.	Privacy and Security
Organizational Responsibilities	Transparency	Platforms should clearly inform users how their data is collected, used and shared.	Accountability
Organizational Responsibilities	Ethical data management	Prevents personal data from being used for manipulation, unfair profiling or unauthorized sharing.	Justice and Accountability

Table 6.1 Proposed Solutions for Data Privacy Issues

CHAPTER 7 CONCLUSION AND RECOMMENDATIONS

7.1 CONCLUSION

In conclusion, data privacy is an important ethical issue in social media platforms because these platforms collect, store and use large amounts of personal information from users. Social media provides many benefits, such as communication, entertainment, education and business promotion. However, it also creates serious privacy risks when user data is collected without clear consent, shared with third parties or used for targeted advertising and profiling.

This report has discussed the ethical concerns of data privacy through privacy, security, justice and accountability. Privacy issues happen when users do not have enough control over their personal information. Security issues happen when platforms fail to protect data from hackers, cyberattacks and unauthorized access. Justice issues happen when data use creates unfair treatment, manipulation or unequal protection for vulnerable users. Accountability issues happen when it is unclear who should be responsible for data misuse or privacy violations.

The study also shows that the real causes of data privacy problems include data-driven business models, lack of user awareness, weak regulatory enforcement and rapid technological development. These causes can lead to serious impacts on individuals, organizations and society. Individuals may face

identity theft, emotional distress and loss of privacy. Organizations may face legal penalties, financial loss and damage to reputation. Society may experience digital surveillance, reduced public trust and manipulation through targeted information.

Therefore, data privacy problems cannot be solved by one party only. Social media companies, governments, users and third parties must work together to protect personal information. Responsible data governance, ethical computing practices and stronger security measures are necessary to make social media safer and more trustworthy.

7.2 RECOMMENDATIONS

This report has identified several recommendations with respect to improving online privacy protection on social networking sites.

1. Improve online privacy education. Users need to know how social networks collect, use and share personal data. Schools, universities, governments and businesses should create awareness campaigns to educate consumers about the dangers of phishing scams and how to properly set their privacy settings when using social media platforms.
2. Enhance laws governing online privacy protection. Governments should update their data protection laws to include the new technologies of artificial intelligence; big data; and algorithmic profiling. Stricter regulations will add another layer of assurance that companies operating social websites follow ethical and legal guidelines while handling user data.
3. Enhance transparency of social networking sites. Social networking companies should write their privacy policy in plain and simple terms that are

easy to read and understand. Users should know, at a minimum, what data is collected; how it is collected; why it is collected; and who has access to this information.

4. Businesses should invest more money into cybersecurity. Companies that operate social networking sites should implement strong cybersecurity controls, including, but not limited to, encryption; multi-factor authentication; regular security testing; and data minimization. These steps can decrease the potential for data breaches and/or unauthorized access to personal user information.

5. Encourage ethical computing practices. Businesses and developers of social networking sites should consider the ethical implications of their computing practices at all times.

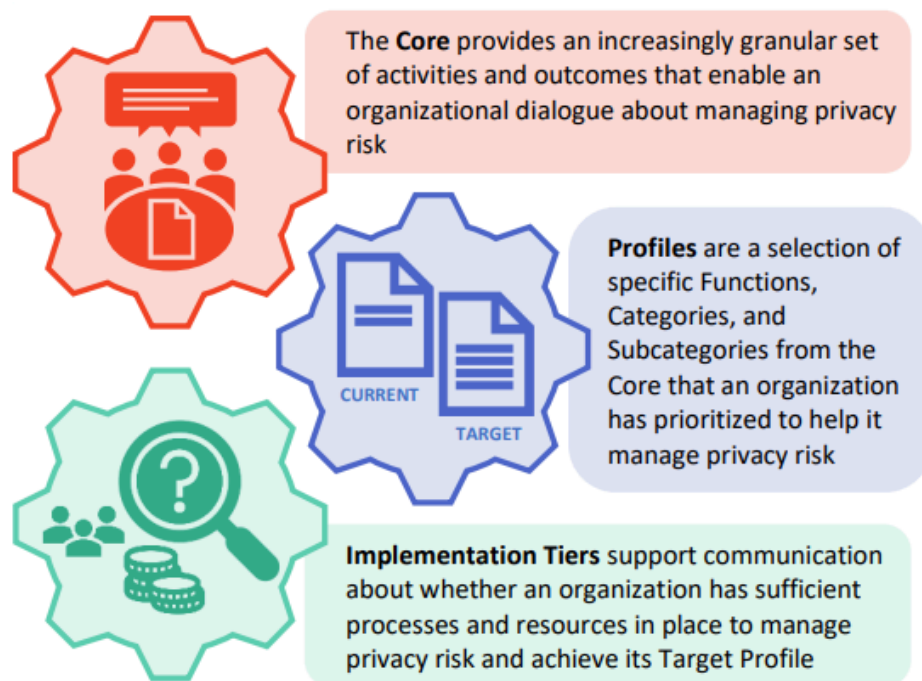


Figure 7.1 Privacy Risk Management Framework

Source: National Institute of Standards and Technology (NIST), Privacy Framework. [12]

To close, user privacy is a personal responsibility. Users must not share too much personal information; periodically review their privacy settings; and proceed cautiously when accepting terms and conditions of service. By being mindful of their behavior, users can lessen the risks associated with their own privacy.

In conclusion, individual responsibility towards data privacy exists on social media. There are multiple players that share this responsibility; stronger legislation, improved technology, more transparent organizations and responsible users will all contribute to building a more secure online world. Ethical data management practices need to remain in place in order to allow social media to provide positive contributions to society while protecting users' rights and privacy.

Solution Area	Main Action	Responsible Party	Expected Impact
Technology Solutions	Use multi-factor authentication, encryption, privacy-by-design and data minimization.	Social media platforms and developers	Reduces account hacking, data breaches, identity theft and unauthorized access.
Policy Solutions	Strengthen privacy laws, increase penalties and	Government and regulators	Improves legal accountability and forces companies to

	conduct regular audits.		follow privacy standards.
User Responsibilities	Review privacy settings, limit personal sharing and avoid suspicious links.	Social media users	Reduces oversharing, scams, cyberstalking and personal privacy risks.
Organizational Responsibilities	Increase transparency, manage data ethically and invest in cybersecurity.	Social media companies	Reduces reputational damage, financial loss and loss of public trust.
Ethical Computing Practices	Review algorithms and avoid manipulative or biased data use.	Developers and organizations	Reduces unfair treatment, manipulation and negative social impact.

Table 7.1 Summary of Proposed Solutions and Expected Impact

REFERENCES

- [1] Pew Research Center (2023) – How Americans View Data Privacy
<https://www.pewresearch.org/internet/2023/10/18/how-americans-view-data-privacy/>
- [2] Reuters (2025) – Facebook Privacy Lawsuit
<https://www.reuters.com/legal/government/facebook-must-face-dc-attorney-general-lawsuit-tied-cambridge-analytica-scandal-2025-07-31/>
- [3] Pew Research Center (2023) – Key Findings About Americans and Data Privacy
<https://www.pewresearch.org/short-reads/2023/10/18/key-findings-about-americans-and-data-privacy/>
- [4] TechSpot (2025) – TikTok and Children's Data Collection
<https://www.techspot.com/news/109600-tiktok-accused-collecting-personal-data-thousands-canadian-children.html>
- [5] Ullah, I., Boreli, R., & Kanhere, S. (2020) - Privacy in Targeted Advertising: A Survey
<https://arxiv.org/abs/2009.06861>
- [6] SecurityWeek. (2023). *TikTok faces fresh European privacy investigation over China data transfers*.
<https://www.securityweek.com/tiktok-faces-fresh-european-privacy-investigation-over-china-data-transfers/>

- [7]Pew Research Center (2023) - Views of data privacy risks, personal data and digital privacy laws
<https://www.pewresearch.org/internet/2023/10/18/views-of-data-privacy-risks-personal-data-and-digital-privacy-laws/>
- [8]The Guardian. (2018). *Facebook's week of shame: data breach revelations and public backlash*.
<https://www.theguardian.com/technology/2018/mar/24/facebook-week-of-shame-data-breach-observer-revelations-zuckerberg-silence>
- [9] Cybersecurity and Infrastructure Security Agency (CISA). Multifactor Authentication.
<https://www.cisa.gov/topics/cybersecurity-best-practices/multifactor-authentication>
- [10] European Data Protection Board (EDPB). Principles of Data Protection.
<https://www.edpb.europa.eu/system/files/2023-06/Principles%20Infographic.pdf>
- [11] Federal Trade Commission (FTC). Protecting Personal Information: A Guide for Business.
<https://www.ftc.gov/business-guidance/resources/protecting-personal-information-guide-business>
- [12] National Institute of Standards and Technology (NIST). Privacy Framework.
<https://www.nist.gov/privacy-framework>

LIST OF ATTACHMENTS

Attachment A: Pew Research Center Survey Results

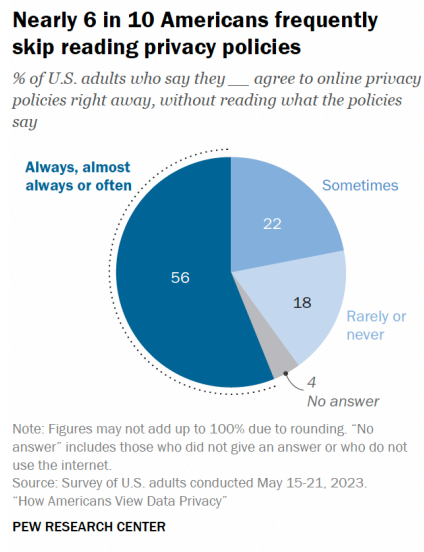
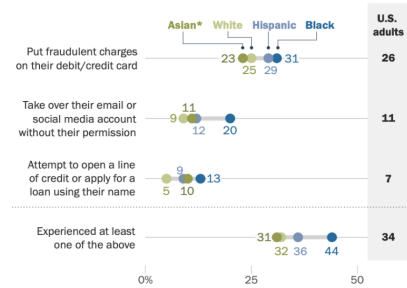


Figure 4.1 Skipped reading privacy policies

Source: Pew Research Center (2023), How Americans View Data Privacy.[1]

Black adults are more likely than other racial and ethnic groups to say they have dealt with an online hack in the last 12 months

% of U.S. adults who say that in the last 12 months, they had someone ...



* Estimates for Asian adults are representative of English speakers only.
 Note: White, Black and Asian adults include those who report being only one race and are not Hispanic. Hispanic adults are of any race. Those who did not give an answer are not shown.
 Source: Survey of U.S. adults conducted May 15-21, 2023.
 "How Americans View Data Privacy"
 PEW RESEARCH CENTER

Figure 4.2 Experienced at least one cyber-related event

Source: Pew Research Center (2023), Key Findings About Americans and Data Privacy.[3]

Most Americans don't trust social media CEOs to handle users' data responsibly

% of U.S. adults who say they have **very little or no trust at all** that leaders of social media companies will ...



Note: Those who did not give an answer or who gave other responses are not shown.
 Source: Survey of U.S. adults conducted May 15-21, 2023.
 "How Americans View Data Privacy"

PEW RESEARCH CENTER

Figure 4.4 Most Americans Do Not Trust Social Media CEOs to Handle Users' Data Responsibly

Source: Pew Research Center (2023), Key Findings About Americans and Data Privacy.[3]

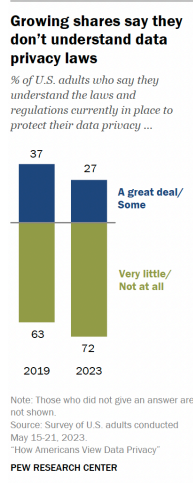


Figure 5.1.4

Growing shares say they dont understand data privacy laws

Source: Pew Research Center (2023) - Views of data privacy risks, personal data and digital privacy laws [7]

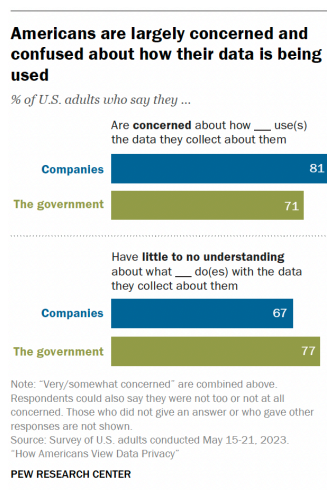


Figure 4.4 Americans are largely concerned and confused about how their data is being used

Source: Pew Research Center (2023), *How Americans View Data Privacy*. [1]

Attachment B: Cambridge Analytica Supporting Documents

Facebook must face DC attorney general's lawsuit tied to Cambridge Analytica scandal

By Mike Scarcella

August 1, 2025 3:58 AM GMT+8 · Updated August 1, 2025



A keyboard is placed in front of a displayed Facebook logo in this illustration taken February 21, 2023. REUTERS/Dado Ruvic/Illustration/File Photo [Purchase Licensing Rights](#)

Attachment C: TikTok Privacy Investigation Supporting Documents

TECHSPOT

LOGIN

TRENDINGREVIEWSFEATURESDOWNLOADSTHE BESTTECH DBPRODUCT FINDERFORUMS

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By Skye Jacoby · September 24, 2025 at 11:20 AM | 15 comments



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